

Temporal Dynamics of Fronto-Parietal Functional Connectivity Orchestrated through Theta Oscillations During Increasing Working Memory Load

Abstract

This study investigated whether theta-band (4-8Hz) fronto-parietal functional connectivity follows an inverted U-shaped trajectory as a function of working memory load. Using an existing EEG dataset from 65 healthy participants performing an auditory digit span task (lengths: 5, 9 or 13 digits), we computed the debiased weighted Phase Lag Index (dwPLI) during the 2-second inter-stimulus encoding interval. To formally test the inverted U-shape hypothesis, a two-lines approach via a generalized linear mixed effects model (GzLMM) was applied. A breakpoint was specified at serial position 6 to estimate and compare slopes before and after this hypothesized capacity limit. Results revealed no evidence for an inverted U-shape relationship, as neither slope reached statistical significance and only the low load slope showed a trend in the hypothesized direction. In conclusion, theta-band fronto-parietal dwPLI did not vary systematically with WM load. The null results, limitations and future directions are discussed in this manuscript.

Introduction

A core cognitive phenomenon, the working memory (WM), is responsible for the temporary encoding, maintenance, manipulation and retrieval of small amounts of information in a readily accessible manner for a broad range of cognitive tasks (Baddeley, 1992; Cowan, 2013; Gulbinaite et al., 2014; Leung et al., 2004; Otstavnov, 2024). The WM has a limited storage and duration capacity, which can be overloaded, resulting in the declination of cognitive performance on a variety of tasks (Chikhi et al., 2022; Moko et al., 2023). For example, WM overload can increase error-rates, response times and susceptibility to distractions which can have detrimental consequences in real-life occupations such as control room operators, paramedics or construction workers (Ghalenoei et al., 2021; Pooladvand & Hasanzadeh, 2022; Zaphir et al., 2024). Furthermore, WM overload can negatively affect psychological wellbeing and emotion regulation, which in turn, may further impair WM performance (Adamczyk et al., 2022; Van Dijk et al., 2020). For these reasons, it is important to investigate the underlying neural mechanisms of WM processes.

These neural mechanisms are thought to be organized through a fronto-parietal network which encapsulates at least two crucial systems: (a) the frontal executive system which directs cognitive resources and attention to control what is being represented and maintained in the WM, and (b) the parietal representational system which represents and maintains the WM content (Berger, 2022; Ratcliffe, 2022). One mechanism that is thought to be crucial in facilitating the frontal executive and parietal representational systems to communicate efficiently, is the oscillatory activity in the theta-band (4-8 Hz) as observed through electro-encephalography (EEG) studies (e.g., Riddle et al., 2020; Zhang et al., 2015). For example, a meta-analysis by Chikhi et al. (2022) found that frontal theta power increases with increasing WM load, highlighting its role during encoding and maintenance. Additionally, studies focusing on the power increases during the maintenance period found the largest effect sizes, suggesting that this phase is the critical period for theta involvement. Therefore, theta oscillations have been proposed to facilitate the encoding of new WM information while maintaining existing WM content through the regulation of fronto-parietal interactions (Ratcliffe, 2022).

To further uncover the role of theta oscillations in the WM system, Kosachenko et al. (2023) exploited an important characteristic of the WM: its limited capacity (Cowan, 2001; Cowan, 2010; Zhang et al., 2015). The authors employed EEG power analysis in combination with an auditory digit span task (1-13 digits) in which subjects had to encode, maintain and recall strings of numbers. These strings contained 5, 9, or 13 digits reflecting low-, moderate-, and high levels of working memory load, respectively. They found that theta oscillations at the frontal midline electrode (Fz) could track different degrees of WM load. Specifically, as working memory load increased, the power of the theta frequency band increased. However, when the WM was loaded at capacity (i.e., 7-9 digits), the theta oscillations' power stayed at a plateau, after which it started decreasing as the WM got overloaded, forming an inverted U-shape pattern. This triphasic pattern of activity suggests that effective WM may depend on frontal theta activity.

Importantly, the study by Kosachenko et al. (2023) has several shortcomings. First, they did not formally test for an inverted U-shape, they simply established it through visual inspection. Second, they restricted their analysis to a single electrode, while the WM network is distributed across the frontal and parietal brain regions. Finally, they did not fully take into account the communicative role of theta oscillations as they only investigated power shifts, while functional connectivity may be more fitting to assess the role of theta oscillations in WM (overload) (Chikhi et al., 2022).

Indeed, functional connectivity (i.e., statistical dependency between regions) has proven to be fruitful in revealing the neural mechanisms of the fronto-parietal WM network, and its response when capacity limits are approached or even, exceeded. For instance, Zhang et al. (2015) studied the relationship between theta band (4-8 Hz) functional connectivity and WM load. They recorded EEG from 32 electrodes, while sixteen participants performed a visual WM task requiring them to memorize sequences of pictures ranging from one to six items. Functional connectivity was operationalized through the directed transfer function (DTF) during the delay period (i.e., the three second period after encoding) and they found that fronto-frontal and fronto-parietal functional connectivity strength increased with increasing load until the capacity limit of four items was reached, after which it declined. These findings more directly support the idea that overload leads to a breakdown of communication in the fronto-parietal WM network instead of a simple decrease of power (Kosachenko et al., 2023).

However, some limitations should still be noted. First, the researchers relied on a small sample size ($N=16$), and a WM task with relatively modest demands to establish overload. Second, the study investigated visual WM, so considering that modality-specific factors can influence WM processes, generalizing the results of Zhang et al. (2015) to auditory WM is ill-advised (Saults & Cowan, 2007). Third, similar to Kosachenko et al. (2023), Zhang et al. (2015) lacked an appropriate test for inverted U-shapes. Instead, they relied on one-way ANOVAs that can identify localized differences but fail to model the global quadratic trend that is hypothesized (Haans et al., 2016; Simonsohn, 2018). Fourth, the researchers used a suboptimal functional connectivity technique. Specifically, they used DTF. Although this technique has some benefits like providing information on directionality of connectivity, it has some important drawbacks as well: it fits a multivariate autoregressive model to the data, meaning that it does not compute phase connectivity directly from raw signals and it is highly dependent on the adequacy of model assumptions and choices (Chiarion et al., 2023; Kaminski & Blinowska, 2017). Furthermore, it is sensitive to volume conduction (Haufe et al., 2011; Liu et al., 2024; Shafiei et al., 2024).

Taken together, it remains an open question whether the results by Zhang et al. (2015) generalize to larger samples, more demanding WM tasks, other modalities, other functional connectivity techniques, or to a formal statistical test of the inverted U-shape.

Current Project

The current study aims to assess the research question of whether theta band functional connectivity between frontal and parietal electrodes increases (low WM load), plateaus (WM loaded at capacity) and decreases (WM overload) as the result of an increasing auditory WM (over)load. Using EEG data of 65 participants performing an auditory WM digit span task, debiased weighted Phase Lag Indices (dwPLI) were computed and a formal test of the proposed inverted U-shape was performed using a generalized Linear Mixed Model (gzLMM).

Method

Dataset and sample size

The current study used the open-access EEG dataset provided by Pavlov et al. (2024) as described in Pavlov et al. (2022), approved by the Ural Federal University ethics committee. This dataset captures the neural and physiological correlates of sequential auditory memory encoding, maintenance and retrieval under varying cognitive load conditions. It includes

electroencephalographic (EEG) data, electrocardiographic (ECG) data, photoplethysmographic (PPG) data, pupillometry data, and behavioral data of 86 participants performing a digit span task. The behavioral data contained trial-by-trial and item-by-item recall responses, manually transcribed from recorded speech.

The present study draws on the EEG and behavioral data only. Importantly, due to technical issues, 21 subjects lack EEG data in this dataset, resulting in a final sample of 65 participants (58 females, 7 males) with ages ranging from 18 to 37 ($M = 20.00$, $SD = 2.60$). One participant (ID = 64, age = 18, female) had a large number of artifactual epochs and was therefore excluded from further analyses. Hence, the final data set consisted of 64 subjects. All participants had normal or corrected-to-normal vision and no history of neurological, psychiatric, or hearing disorders.

EEG recording and equipment

EEG data were recorded at a sampling rate of 1000 Hz using a 64-channel EEG system (ActiCHamp, Brain Products, Germany) with electrodes arranged according to the extended 10–20 mapping. The FCz channel served as the online reference, and the Fpz channel served as the ground electrode. Because the current project specifically focuses on frontal- and parietal theta-band activity, the following channels were selected for analysis: 'Fz', 'F1', 'F2', 'Pz', 'P1', 'P2'. Especially the frontal midline electrode (i.e. 'Fz') and the parietal 'Pz' channel are often investigated in the context of theta oscillations during working memory processes (e.g., Berger & Sauseng, 2022; Hsieh & Ranganath, 2013; Nakamura-palacios et al., 2023; Pusztá, 2022; Saliassi et al., 2013; Toppi et al., 2018). To this selection, we added 'F1/2' and 'P1/2' for a more comprehensive estimation of frontal and parietal theta-activity, respectively. This selection overlaps with channels used in other studies investigating fronto-parietal connectivity and WM processes, where electrode clusters are used rather than single electrodes (e.g. Chang et al., 2023; Erickson et al., 2019).

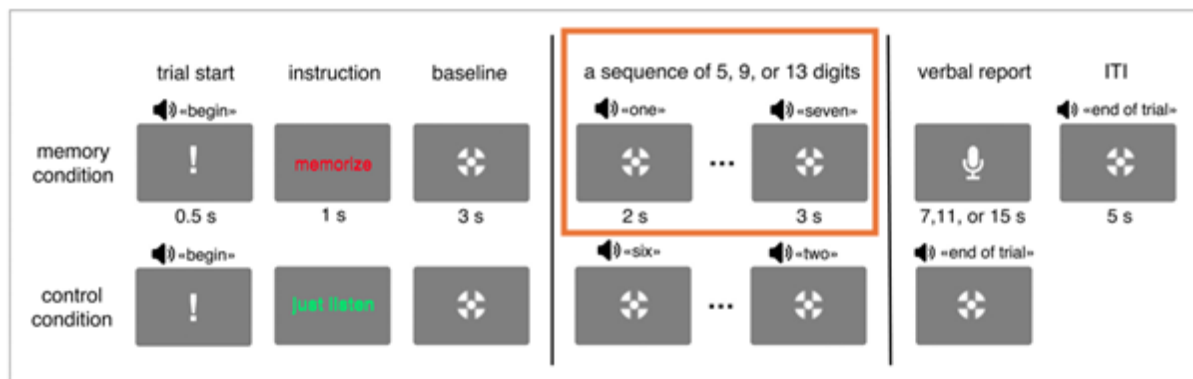
Procedure and Paradigm

To assess auditory WM, participants performed an auditory digit span task where participants encoded, maintained, and recalled strings of either 5, 9 or 13 auditory presented digits. These represented low, moderate or high levels of WM load, respectively. Additionally, to ensure constant visual input, participants continuously focused on a fixation cross throughout the experiment, presented on a monitor (80cm from the participant) flanked by two speakers.

The experiment consisted of nine blocks, with 108 active recall trials and 54 passive listening (control) trials. Each block followed a set structure: three control trials (one per load level), followed by twelve randomized active recall trials (four per load level), and concluding with three additional control trials. The control trials were introduced to dissociate any WM processes found by analyzing the data from passive auditory perception. Each trial began with the voice command “begin” and an instruction specifying the condition (memory or passive listening). Following a 3-second baseline, the digit sequence was presented at 70 dB with a 2-second inter-stimulus interval (ISI). Following the final digit, a 3-second retention interval was provided. In the memory condition, participants then verbally recalled the digits in serial order; the recall phase was timed at 7, 11 or 15 seconds for the 5, 9, or 13 digit loads, respectively. In the control condition, participants were not required to recall the digits, and the retention interval led directly to a 5-second inter-trial interval. Figure 1 shows an overview of the experimental paradigm (Kosachenko et al., 2023, p. 2).

Figure 1

Visualization of the Experimental Paradigm



Note. Adapted from Kosachenko et al., 2023. EEG and Pupillometric Signatures of Working Memory Overload, by Kosachenko et al., 2023, *Psychophysiology*, 60(6), p. 2 (<https://doi.org/10.1111/psyp.14275>). CC BY-4.0. The orange rectangle indicates the encoding and maintenance phase, in which data was epoched to the individual digit-encoding period following presentation offset.

Data Preprocessing

Data preprocessing and subsequent phase-connectivity analyses were performed using the MNE-python toolbox (Gramfort et al., 2013). The continuous EEG signal was first broadband filtered (1-40 Hz) to remove slow drifts, line noise, and high frequency muscle artifacts. Electrodes identified as containing excessive noise during visual inspection were interpolated using spherical spline interpolation. Following interpolation, the data were re-referenced to the common average. To isolate specific digit-encoding activity, the continuous data were segmented into epochs from 300 ms to 2000ms relative to the onset of each digit presentation. Note that the first 300 ms of a trial were excluded because of the overlap with auditory digit presentation.

Artifact rejection was conducted in two sequential stages: (1) manual visual rejection of epochs containing prominent, non-stereotypical artifacts and (2) Independent Component Analysis (ICA) was fitted on the remaining epochs using the Extended Infomax algorithm via `max_iter = 'auto'`, to isolate and remove artifacts related to ocular activity (e.g. blinks and lateral movements). To avoid over-fitting and account for rank deficiency, the number of fitted ICA components was dynamically adjusted for each participant by subtracting the number of interpolated bad channels and one additional degree of freedom sacrificed during the common average referencing ($N_{\text{channels}} - N_{\text{interpolated}} - 1$). Cleaned epochs were then exported for downstream phase synchronization analysis.

Fronto-parietal theta connectivity analysis: *debiased wPLI*

EEG connectivity analysis was performed using custom Python scripts employing the SciPy library (Virtanen et al., 2020). To quantify theta-band phase synchronization between frontal (Fz, F1, F2) and parietal (Pz, P1, P2) regions during WM encoding, the cleaned epochs were bandpass-filtered to the theta range (4-8Hz) using a zero-phase finite impulse response (FIR) filter to prevent phase distortion. Real-valued theta signals were then converted into complex analytic signals via the Hilbert transform to extract instantaneous phase values. Functional connectivity (i.e., debiased weighted Phase-Lag Index, see later) was evaluated epoch-by-epoch (300-2000ms post-digit onset) across all nine pairwise combinations of the three frontal and three parietal channels.

A commonly used phase-based connectivity measure is the phase-locking value (PLV). This measure captures phase angle consistency, but is highly susceptible to volume conduction (Cohen, 2014). As volume conduction typically arrives at different sensors simultaneously with near-zero time

delays, a measure like the Phase-Lag Index (PLI) decreases sensitivity to volume conduction by discarding phase lags centered around 0 or π . The weighted PLI (wPLI) further extends and improves the PLI framework by weighting observations by the magnitude of the imaginary component of the cross-spectrum. Consequently, phase differences closer to ninety degrees exert the most influence on the final estimate, while those near zero are down-weighted, thereby increasing resistance to noise and volume conduction (Samiei et al., 2022; Shafiei et al., 2024). However, both PLI and wPLI suffer from a positive sample-size bias in small datasets and short epochs. This bias arises from the inclusion of epoch self-pairings in the standard computation (Vinck et al., 2011). To correct this, we utilized the debiased wPLI (dwPLI), which excludes self-pairings and is formally defined as:

$$\text{debiased wPLI} = \frac{\sum_{j=1}^N \sum_{k \neq j} \Im\{X_j\} \Im\{X_k\}}{\sum_{j=1}^N \sum_{k \neq j} |\Im\{X_j\} \Im\{X_k\}|}$$

Where $\Im\{X_j\}$ represents the imaginary part of the cross-spectrum between frontal and parietal channels at observation j (Vinck et al., 2011). Finally, for each epoch, a single scalar value representing global fronto-parietal theta connectivity was computed by averaging the dwPLI across all pairwise combinations of frontal and parietal channels. These epoch-level connectivity values were compiled alongside behavioral metadata (digit position in a sequence, sequence length and recall accuracy).

Statistical Analyses

All statistical analyses were conducted in R (Version 4.2.2; R Core Team, 2022) using dplyr and tidyr (e.g., demographic summary statistics, data manipulation and cleaning) (Wickham et al., 2014a, 2014b), ggplot2 (e.g., generation of figures) (Wickham, 2016), lme4 (e.g., model construction and fitting) (Bates et al., 2015), performance (e.g., model assumptions check) (Lüdtke et al., 2019), multcomp (e.g., general linear hypotheses testing) (Hothorn et al., 2008).

Two-Lines Procedure

The current study attempts to test whether fronto-parietal theta connectivity (i.e., dwPLI) changes as a function of increasing WM load (i.e., position of a digit in a sequence from 1 to 13). Specifically, we test whether this relationship follows an inverted U-shape, with a plateau near the WM capacity limit and a decline once the capacity is exceeded. The traditional methods of evaluating the presence of an inverted U-shape relationship are known as polynomial/quadratic regression frameworks. However, as Simonsohn (2018) suggests, such techniques are accompanied with high false-positive rates and plateau-driven artefacts, failing to explicitly assess the presence of an inverted U-shape. Therefore, we adapted the “Two-Lines procedure” as proposed by Simonsohn (2018).

The core logic of the Two-Lines framework involves an interrupted regression to determine whether the relationship between WM load (X) and connectivity (Y) is significantly positive up to a specific critical inflection threshold or breakpoint (x_c), after which the relationship significantly reverses and becomes negative. This breakpoint should in theory be the point where WM reaches its limit. While the original two-lines procedure by Simonsohn (2018) was developed for independent observations, the hierarchical structure of our EEG dataset (multiple within-subject epochs) requires accounting for within-subject dependencies. We therefore extended the interrupted regression framework into a segmented or piecewise linear mixed-effects model (Benwell et al., 2019; Yu et al.,

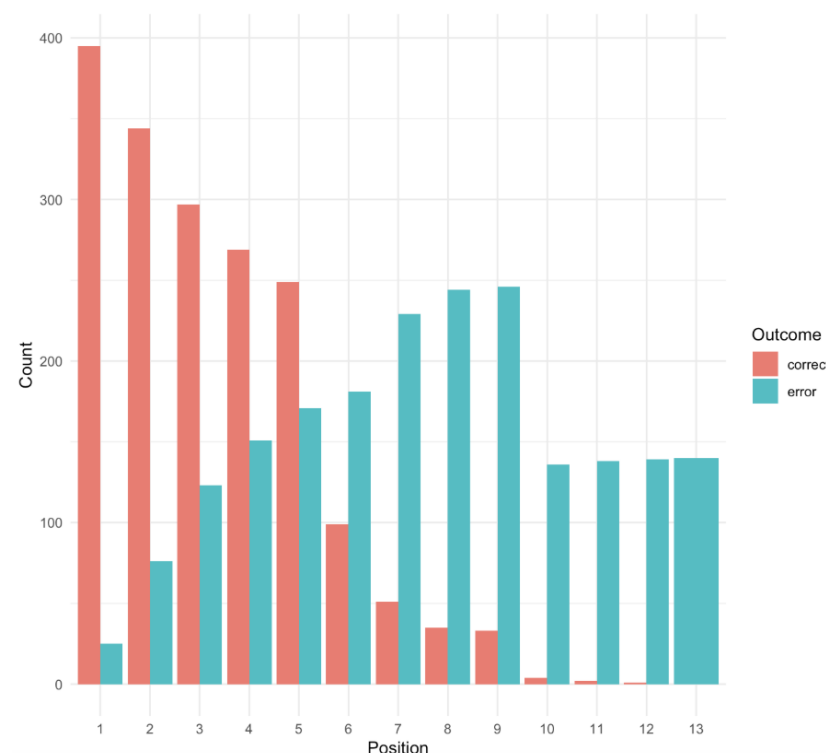
2021). This model estimates separate linear slopes before and after a designated breakpoint (x_c), while accounting for baseline inter-individual differences in connectivity via random intercepts.

Breakpoint Determination

Empirically defining a precise and absolute breakpoint (x_c) at which WM encoding capacity maxes out remains a challenge, due to inconsistency in the observed reported limits. For instance, visual WM tasks often identify a capacity limit around a load level of 4 (Zhang & Luck, 2008), whereas sequential auditory/digit tasks have observed theta-band plateaus or inflections at higher loads between levels 6 and 9 (Kosachenko et al., 2023). Therefore, a data-based approach was adopted in the current project. Specifically, the recall accuracy of digits, represented across each position in the sequence (1-13), was inspected to determine the first digit position at which error counts exceeded correct-response counts, reflecting the point at which task demands on average appeared to surpass participants' WM capacity. Based on this criterion, a breakpoint was set at serial position 6. Figure 2 displays grouped bar plots showing the number of correct and incorrect responses at each serial position.

Figure 2

Accuracy Counts by Position



Note. A visual depiction of the breakpoint selection criterion, whereby serial position 6 is the point at which the number of incorrect responses becomes greater than the number of correct responses.

Piecewise Predictor Construction

Following breakpoint selection, two predictors (load 1 and load 2) were constructed to implement the two-lines model. Load 1 represents the linear increase in WM load up to the breakpoint. This predictor increased linearly from serial position 1 to the breakpoint ($x_c = 6$) and then remained constant at 6 for all following positions. Load 2 represented the post-breakpoint change in slope. This predictor was coded as 0 before the breakpoint and increased linearly from 1 to 7 for

positions after the breakpoint, so that load 1 + load 2 equals the exact position in the sequence. This parametrization creates a hinge function at the breakpoint, allowing separate estimation of pre-breakpoint and post-breakpoint slopes within a single regression model.

Model Fitting

A linear mixed-effects model (LMM) with dwPLI as dependent variable, Load 1 and Load 2 as fixed effects and a participant-specific intercept as random effect was initially planned (Benwell et al., 2019; Simonsohn, 2018; Yu et al., 2021). For a given trial i nested within participant j , this model was specified as:

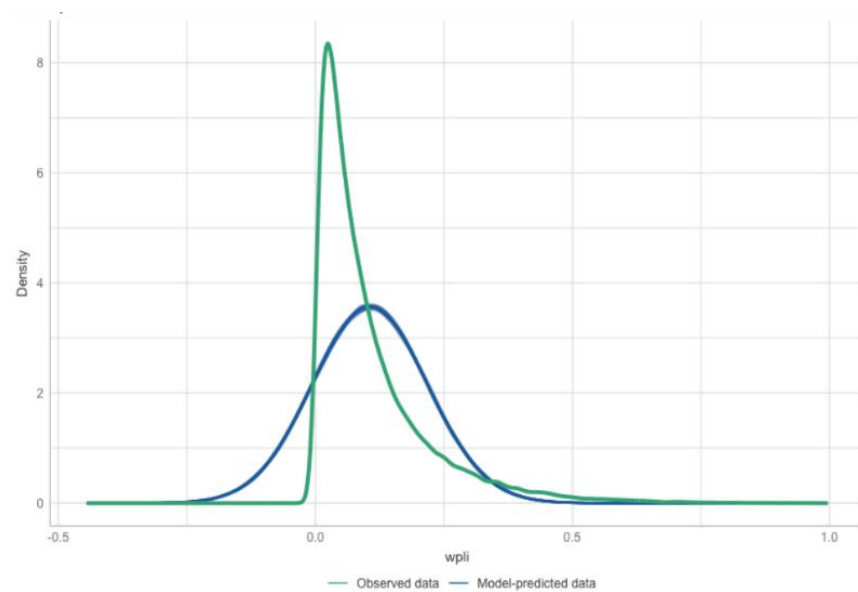
$$wPLI_{ij} = \beta_0 + \beta_1(\text{Load1}_{ij}) + \beta_2(\text{Load2}_{ij}) + u_{0j} + e_{ij}$$

Where β_0 represents the fixed-effect intercept, β_1 represents the slope before the breakpoint, β_2 represents the change in slope after the breakpoint, u_{0j} denotes the random intercept for participant j , and e_{ij} represents residual error. To statistically reject the null hypothesis in favor of a true inverted U-shaped relationship, two conditions must be simultaneously met: the initial slope must be significantly positive ($\beta_1 > 0$) and the downstream slope must be significantly negative ($\beta_1 + \beta_2 < 0$), with both fulfilling a two-tailed significance threshold (Simonsohn, 2018).

However, diagnostic assessment of this LMM revealed that while the assumption of homoscedasticity and normality of random effects were largely met, the model severely violated the assumption of normality of residuals and linearity of the relationship between predictors and outcome variable (see Appendix A). This violation is to be expected when working with data that is inherently right-skewed and lower bounded by zero, as is the case with wPLI data (values between 0 and 1). The bad fit of the model was further expressed in the posterior predictive plot, exhibiting a major inconsistency between the model prediction and the observed data (see Figure 3).

Figure 3

Posterior Predictive Check



To address this bad fit of the LMM, a generalized linear mixed model (GzLMM) was fitted using a Gamma error distribution and a log link function. This model explicitly accommodates the

positive skew and accounts for residual variance that scales proportionally with the square of the mean (Ng & Cribbie, 2017). To address nonlinearity, the model implements a log link function, linearizing its relationship with the categorical predictors. Diagnostics evaluations of the assumptions (i.e., positive dependent variable, linearity via Log Link, variance proportional to the square of the means, approximation of normality of residuals) were conducted and satisfied. Therefore, the GzLMM framework was adopted for all subsequent hypothesis testing.

Hypothesis Testing

The primary hypotheses concerned the direction and significance of the slopes before and after the breakpoint. First, we tested whether the pre-breakpoint slope differed significantly from zero and whether it showed a positive trend, consistent with increasing connectivity as WM load increased. Second, we tested whether the post-breakpoint slope differed significantly from zero. The post-breakpoint slope was computed as:

$$PostSlope = \beta_1 + \beta_2$$

These hypotheses were formally assessed through bonferroni-corrected simultaneous tests of general linear hypotheses ($\alpha = 0.05$) using multcomp (Hothorn et al., 2008).

Passive Versus Active Task Analyses

In the case that a significant U-shape was found for the active task, the complete two-lines analysis procedure would be performed, as described, for the passive listening task as a control analysis. This is a necessary control in order to distinguish between WM-driven and auditory-driven effects (Kosachenko et al., 2023).

Results

Data-Processing

Data of 64 subjects was preprocessed (see Dataset and Sample Size), for which 1458 trials were recorded per subject. For one subject (ID = 94), 162 trials lacked a condition label in the events file; these could not be processed by the labelling script and were excluded prior to epoching. As a result, only 1296 trials were recorded for this subject. Across all 64 subjects, visual inspection of the epochs led to the removal of a small proportion (0.88%) of all epochs; a total of 92328 remained from the initial 93150 epochs. Epochs were rejected primarily due to high-frequency noise and drift artifacts. The number of epochs removed per subject ranged from 0 to 116 ($M = 12.84$). Then, for 14 of 64 subjects (22%), bad channel interpolation was necessary. For these subjects, the number of bad channels ranged from 1 to maximally 3 channels. Independent component analysis was used to remove eye-blink artifacts based on topography and waveform. Between 1 and 3 components were removed per subject. After preprocessing, the final dataset consisted of 61791 memory trials (66.93%) and 30537 control trials (33.07%).

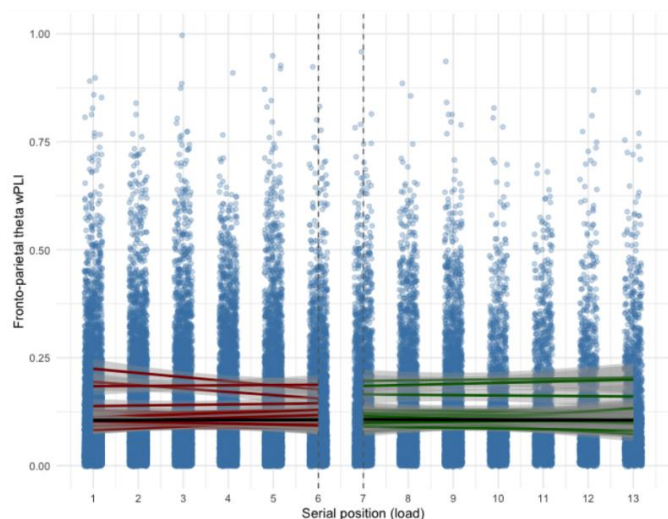
Descriptive Statistics

An overview of the mean, standard deviation (SD), median and Inter-Quartile Range (IQR) of dwPLIs across serial positions (1-13) is shown in table 1. Importantly, the mean and median dwPLI remains rather constant across serial positions. Additionally, note that the number of datapoints (N) decreases per serial position.

Table 1*DWPLI Descriptive Statistics per Digit Serial Position*

Position	N	Mean	SD	Median	IQR
1	6870	0.106	0.111	0.069	0.111
2	6869	0.106	0.110	0.070	0.108
3	6869	0.106	0.112	0.069	0.107
4	6869	0.106	0.109	0.069	0.110
5	6869	0.106	0.113	0.068	0.109
6	4576	0.108	0.112	0.070	0.111
7	4576	0.106	0.111	0.070	0.107
8	4576	0.107	0.112	0.069	0.111
9	4575	0.108	0.114	0.070	0.117
10	2287	0.108	0.111	0.071	0.110
11	2286	0.107	0.111	0.069	0.110
12	2285	0.104	0.111	0.065	0.105
13	2284	0.108	0.114	0.069	0.108

Figure 4 shows the average dwPLI at subject-level, in red for pre-breakpoint and in green for post-breakpoint. Additionally, the grand average across subjects is shown in black. In line with the descriptive statistics reported above, this visualization shows a rather constant dwPLI trajectory across digit serial position.

Figure 4*Weighted Phase Lag Index across Serial Position by Subject*

Note. The red and green lines represent the subject-level mean dwPLI across serial position before and after the breakpoint. The black line is the grand average. Every blue dot is an individual datapoint.

Formal Test of the Predicted Inverted U-Shape

To formally test the predicted inverted U-shape of the relationship between serial positions of digits (proxy for WM load) and fronto-parietal theta dwPLI (proxy for connectivity), a gamma log-link GzLMM was fit predicting the dwPLIs using two re-parameterized slope predictors (Load 1 = pre-breakpoint slope, Load 2 = change in slope post-breakpoint), with the breakpoint (BP) fixed at 6 and a random intercept per subject. Furthermore, two bonferroni-corrected simultaneous tests of general linear hypotheses were performed: one to assess the effect of load 1 (pre-BP) and one to assess the combined effect of load 1 and 2 (post-BP).

In line with the aforementioned descriptive statistics and data visualizations, the effect of serial position of a digit before the BP (load 1) was non-significant ($\beta = 0.0036$, $SE = 0.0024$, $z = 1.478$, $p = 0.279$). When back-transformed from the log scale, this beta coefficient can be interpreted as an increase in wPLI of 0.36% ($e^{\beta_1} - 1 = 0.36\%$) for each unit increase in serial position pre-BP (i.e., going from the first digit in a sequence to the second).

Similarly, the combined effect of load 1 and 2 ($0.003565 - 0.000334 = 0.003231$) was non-significant ($\beta = 0.0032$, $SE = 0.0022$, $z = 1.506$, $p = 0.264$). When back-transformed from the log scale, this combined beta coefficient can be interpreted as an increase in dwPLI of 0.32% (i.e., $e^{\beta_2} - 1 = 0.32\%$) for each unit increase in serial position post-BP. A summary of the results can be found in table 2.

Table 2

Fixed Effects from the Gamma GLMM (Log Link) Predicting wPLI

Predictor	β	SE	z	$Pr(> z)$ adjusted Bonferroni
Load 1	0.003565	0.002412	1.478	0.279
Load 2	-0.000334	0.002174	/	/
Load 1 + 2	0.003231	0.002146	1.506	0.264

Note. Load 2 is not directly interpretable as a standalone effect, as the post-BP slope (Load 1 + Load 2) results are reported in the combined row. Coefficient estimates = β ; Standard Error = SE; z statistic = z; $Pr(>|z|)$ = p value.

Discussion

The current study investigated whether theta-band (4-8 Hz) fronto-parietal FC, operationalized as dwPLI, follows an inverted U-shape trajectory as a function of increasing auditory WM load. This hypothesis resulted from prior work demonstrating triphasic patterns in frontal theta power and connectivity (Zhang et al., 2015; Chikhi et al., 2022; Kosachenko et al., 2023). Using a two-lines statistical approach embedded within a GzLMM, we found no significant effect of auditory WM load on fronto-parietal theta dwPLI, neither pre-BP nor post-BP, and only the pre-BP slope showed the hypothesized positive direction.

Interpretation of the Null-Effect

Power- Related Aspects as Explanation for the Null Finding

The descriptive statistics and visualizations support the interpretation that the null effect reflects a true absence of load-related dwPLI modulation in this dataset, rather than insufficient statistical power. Specifically, mean and median dwPLI values remained remarkably stable across all serial positions (Table 1; Figure 4), with no discernible incline, plateau, or decline. This consistency suggests that fronto-parietal theta phase-based connectivity, as captured by dwPLI, was not sensitive to the WM load manipulations during digit encoding and maintenance as performed in the current paradigm.

However, what should be noted with respect to power is that the task structure automatically creates an asymmetry in observation counts across load levels. Serial positions 1-5 contribute data from all trial types (5-, 9- and 13- digit sequences), whereas positions 10-13 contribute data only from the longest sequences (13- digit sequences). This asymmetry means that estimates of connectivity at higher loads are based on fewer observations and thus estimated with less precision. Although this does not invalidate the performed analysis, it does reduce statistical power precisely where overload effects would be expected to emerge.

Breakpoint Selection and Conceptualization of Overload as Explanation for the Null Finding

The breakpoint (BP, serial position 6) was determined using a data-driven criterion: the first position at which incorrect responses exceeded correct responses. While this approach has face validity (i.e., one would expect that WM overload is accompanied with a strong decrease in accuracy), it may not correspond exactly to the point at which neural correlates of WM (e.g., fronto-parietal theta connectivity) break down. For example, as one of the proposed leading roles of theta based fronto-parietal connectivity for WM is executive control (e.g., Riddle, 2020; Zhang et al., 2015), it may well be possible that while behavioral accuracy already declines, the engagement of the executive control system does not (Moko et al., 2023). Therefore, the relationship between behavioral outcomes and the underlying neural mechanisms is not always straight-forward.

However, since the current analysis uses the BP as a statistical tool for the segmented regression, rather than as a goal to define the exact WM capacity, and considering the unchanging values of the mean wPLI across loads, it is unlikely that a different BP would lead to substantially different results.

Choice of Connectivity Metric as Explanation for the Null Finding

One plausible explanation for the null finding concerns the specific connectivity metric employed. The debiased weighted phase lag index (dwPLI) is a conservative measure designed to minimize spurious connectivity estimates arising from volume conduction and sample-size bias. It achieves this by down-weighting phase differences near zero and π , focusing instead on non-zero lag phase relationships. While this property is methodologically advantageous, it may have obscured functionally relevant connectivity patterns, especially if theta-mediated fronto-parietal coordination during WM operates through in-phase (zero) or anti-phase (π) synchronization.

This possibility finds support in recent literature. For example, Hu et al. (2022) demonstrated that inducing frontoparietal theta synchronization versus desynchronization via transcranial alternating current stimulations significantly affected verbal WM performance. Furthermore, Shafiei et al. (2024) found that combining dwPLI-based and coherence-based connectivity (i.e., connectivity

measure that does not suppress near-zero-lags) yielded superior predictive models of mental workload compared to either metric alone. These findings suggest that zero-lag and near-zero-lag phase relationships, precisely those suppressed by dwPLI, may carry task relevant information about fronto-parietal theta coordination during WM processing.

Individual Differences and Sample Characteristics as Explanation for the Null Finding

Although the current analysis did take into account random intercepts per subject to account for the repeated measures, it looked at the WM load relationship with theta connectivity at a population-level. However, individual differences in executive control abilities, exerted mental effort, and cognitive strategies may substantially moderate this relationship (Chikhi et al., 2022; Zakrzewska & Brzezicka, 2014).

Furthermore, the sample composition requires additional consideration. Participants were predominantly female (57 of 64) and evidence suggests sex differences in both WM performance and associated neural activity. Voyer et al. (2021) reported a small female advantage on auditory digit span tasks, potentially reflecting greater reliance on episodic memory strategies. González-Garrido et al. (2026) found that women exhibit larger frontal theta phase-locking values during visual WM maintenance and may rely on more distributed connectivity networks. If women engage fronto-parietal networks more efficiently or employ different strategies, this could affect where- or whether an inverted U-shape emerges.

Limitations and Future Directions

Several limitations should be acknowledged and may inform future studies. First, the analysis was restricted to six electrode channels (Fz, F1, F2, Pz, P1, P2) which eventually were combined in just one measure of connectivity. This provides a coarse approximation of fronto-parietal connectivity, while inevitably leading to a loss of information. Source-localized connectivity analysis could reveal region-specific effects obscured by scalp-level measurement (Lai et al., 2018; Sanchez-Bornot et al., 2024).

Second, we employed only a single connectivity metric. Because dwPLI discards zero-lag and near-zero-lag phase relationships by design, it cannot capture connectivity dynamics that manifest at these lags. Future studies could employ several complementary connectivity metrics to capture a broader range of connectivity dynamics (Bastos & Schoffelen, 2016; Chiarion et al., 2023). For example, including a metric that preserves zero-lag phase relationships would allow assessment of whether such relationships carry task-relevant information that dwPLI suppresses (Nolte et al., 2004; Stam et al., 2007).

Third, the dwPLI was computed across the full theta band (4-8Hz) rather than at specific frequencies within this range. Given evidence that distinct theta sub-bands may serve different functions, frequency-resolved analyses might reveal load-related modulations not apparent in broader band estimates. For example, Ratcliffe et al. (2022) found that frontal midline theta rhythm slows down as a function of WM load (see also Wolinski et al., 2018).

Finally, the two-lines procedure, while providing a principled test of the inverted U-shape hypothesis, assumes a single breakpoint, constant across subjects. If the true relationship between load and connectivity involves a gradual transition rather than a discrete breakpoint, if the breakpoint varies across individuals or if the real relationship contains more than one breakpoint, this approach may lack sensitivity. For instance, a model proposed by Moko et al. (2023) takes two breakpoints into account. In doing so, they attempt to distinguish between WM overload and cognitive overload, allowing behavioral responses to be affected when WM overload occurs, while not requiring the

systems involved with cognitive control and executive functioning to be affected in the exact same way. In this model it is for example possible that behavioral accuracy decreases, while the subjects are still equally engaged with the task. Future research could further investigate this model and take more individual variability regarding the breakpoint(s) into account.

Conclusion

This study found no evidence that theta-band fronto-parietal functional connectivity, measured via dwPLI, follows an inverted U-shaped trajectory under increasing auditory WM load. The null effect appears genuine rather than reflecting insufficient power, as connectivity values remained stable across all load levels. Several factors may explain this finding: the conservative properties of dwPLI that excludes potentially relevant zero-lag phase relationships, the temporal overlap of encoding and maintenance processes inherent in the digit span paradigm, and the possibility that accuracy-based breakpoint selection does not index the breakdown of neural control mechanisms.

Nevertheless, the findings do not refute the broader hypothesis that fronto-parietal theta coordination supports WM function. Rather, they highlight the importance of methodological choices, particularly the selection of connectivity metrics and the operationalization of overload in detecting load-related neural dynamics. Future research combining multiple connectivity metrics, source-space analysis, and individual-difference approaches may provide a more nuanced understanding of how theta oscillations orchestrate fronto-parietal communication during WM processing.

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AI Acknowledgement

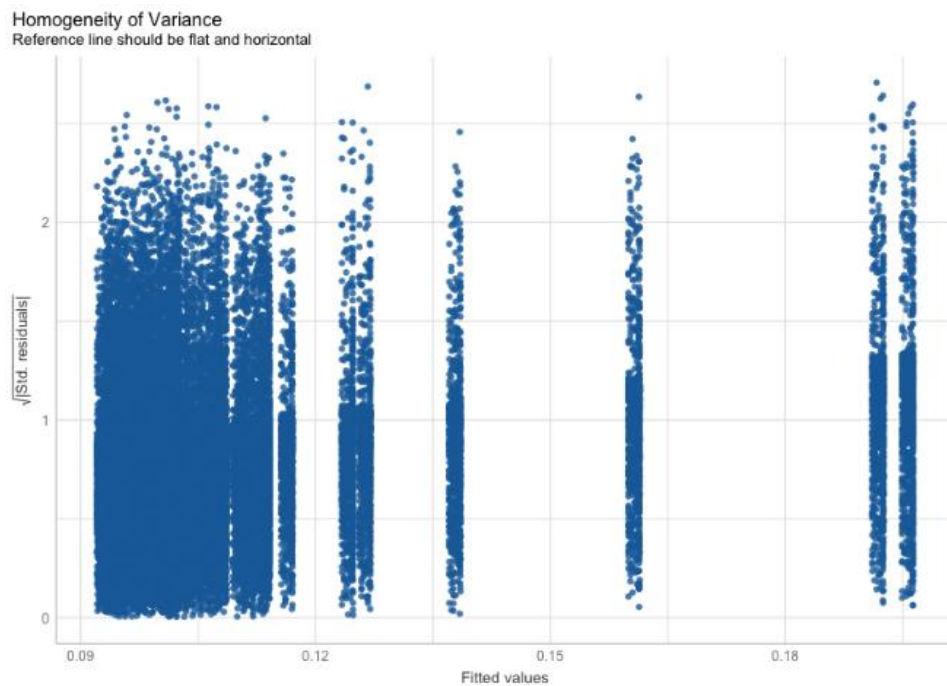
The AI search tool Consensus was used to explore the literature on the topic. For the code of the dwPLI analysis and statistical tests, Claude (Anthropic) helped in the writing of the code, and for troubleshooting and feedback. It also helped to summarize scientific articles and understand certain terms or concepts. With respect to the final paper, Claude and Gemini were used to improve the text flow and coherence by re-phrasing and editing originally written text. AI was not used to generate ideas. All AI output was critically evaluated, and all final text and ideas are the responsibility of the student authors.

Appendix A

Linear Mixed-Effects Model Assumptions

Figure A1

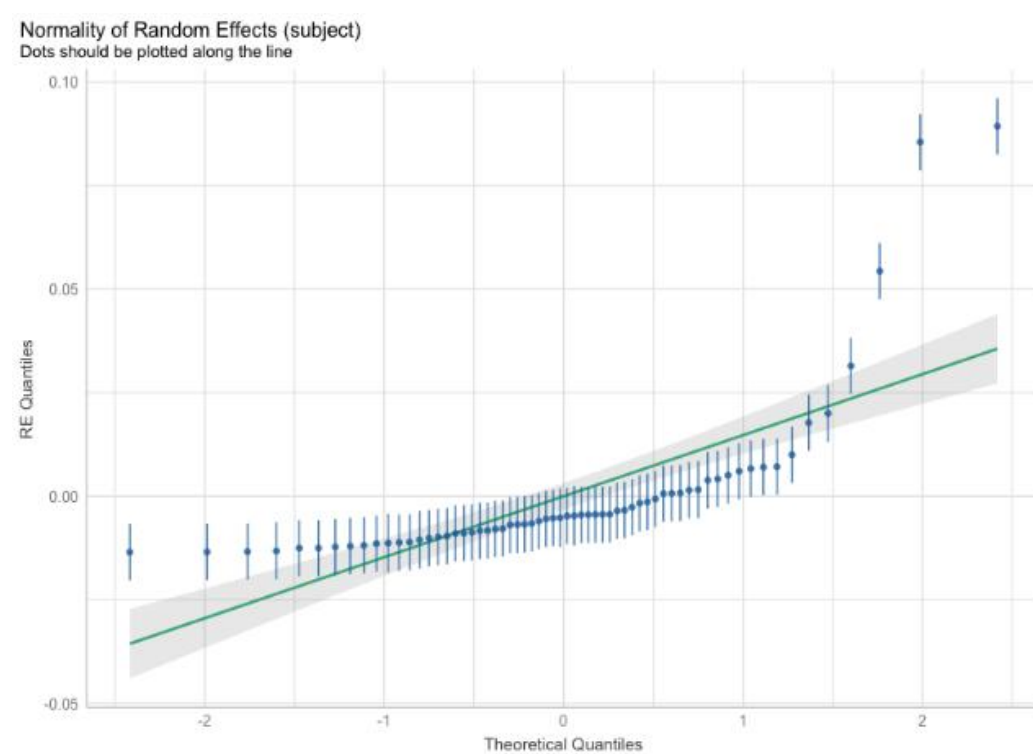
Homogeneity of variance



Note. This figure displays the scale-location plot used to evaluate the assumption of homoscedasticity. The square root of the absolute standardized residuals is plotted against the model's fitted values. The uniform vertical dispersion of data points across the entire continuum of fitted values confirms the assumption of homogeneity of variance.

Figure A2

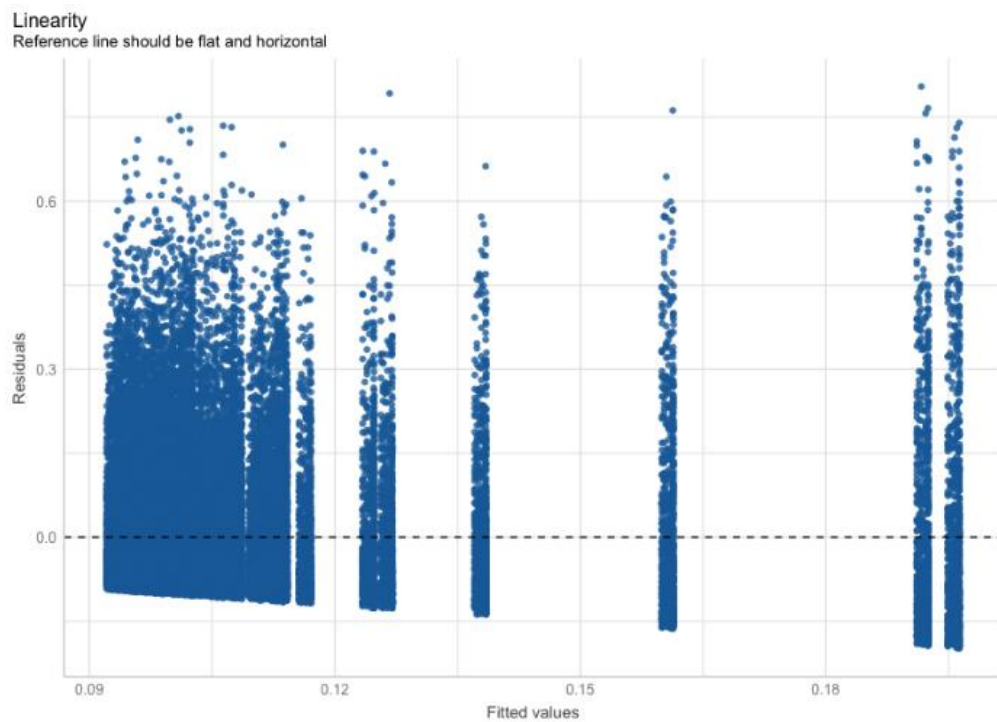
Normality of random effects



Note. This figure displays a quantile-quantile (Q-Q) plot used to assess the assumption of normality for the random effects (subject) within the model. The empirical random effect quantiles are plotted against the theoretical quantiles of a standard normal distribution. The majority of the observations align closely along the reference diagonal.

Figure A3

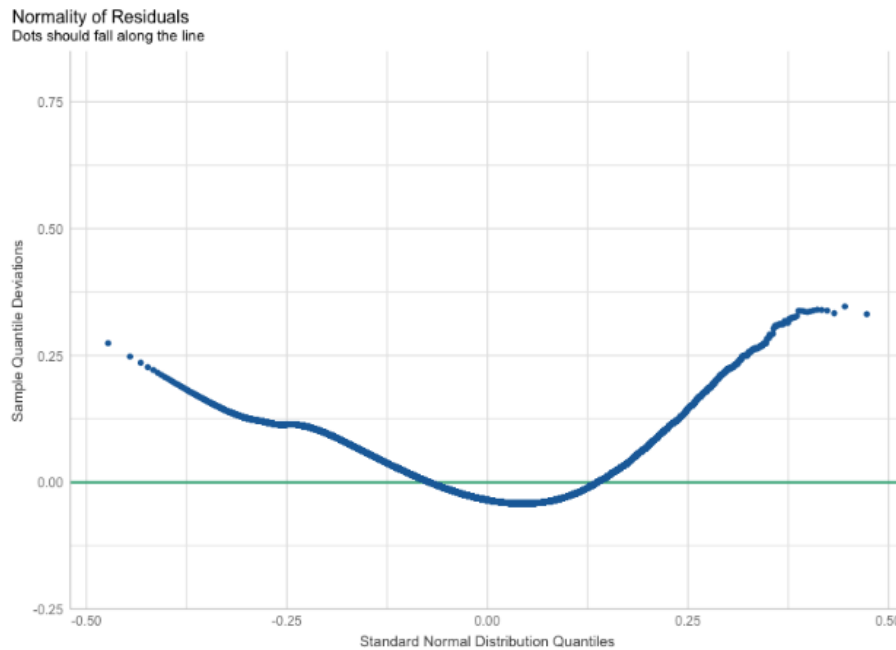
Linearity of the relationship between the predictors and the outcome variable



Note. This figure displays a residuals-versus-fitted plot used to evaluate the assumption of linearity between the predictors and the outcome variable. The raw model residuals are plotted against the predicted fitted values, with a reference line fixed at zero. The distribution exhibits a severe positive skew and a sharp, artificial lower boundary across the entire range of fitted values, which is indicative of a floor effect or truncation in the underlying data.

Figure A4

Normality of residuals



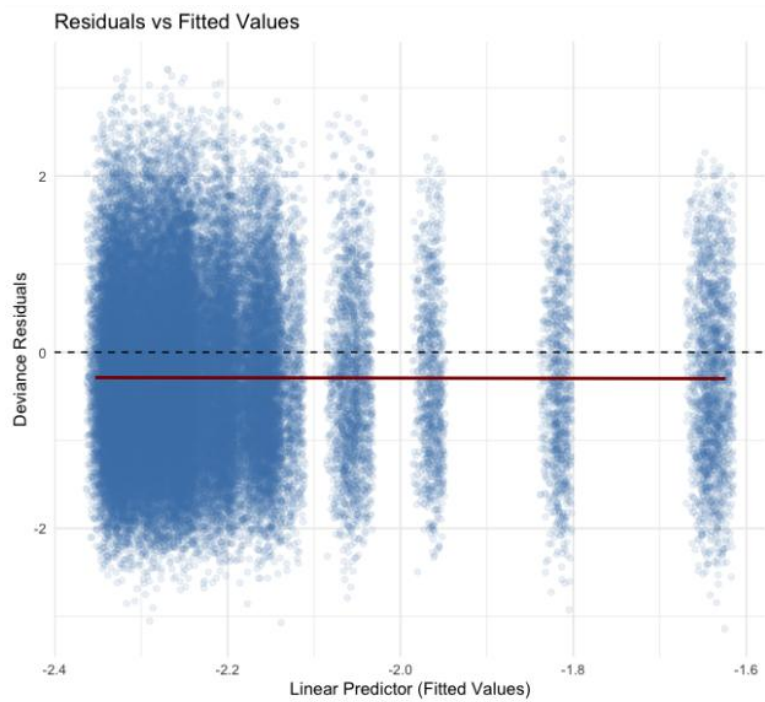
Note. This figure presents a quantile-quantile (Q-Q) plot mapping the sample quantile deviations of the model residuals against the standard normal distribution quantiles. Under the ideal assumption of normality, the observed deviations should fluctuate randomly and closely around the horizontal green reference line at zero. Instead, the distribution exhibits a pronounced, systematic wave-like pattern with substantial departures from the baseline across the entire spectrum. This highly structured, non-random configuration confirms a severe violation of the normality of residuals assumption, indicating that the error terms are heavily skewed and misaligned with a standard normal distribution.

Appendix B

Linear Mixed-Effects Model Assumptions

Figure B1

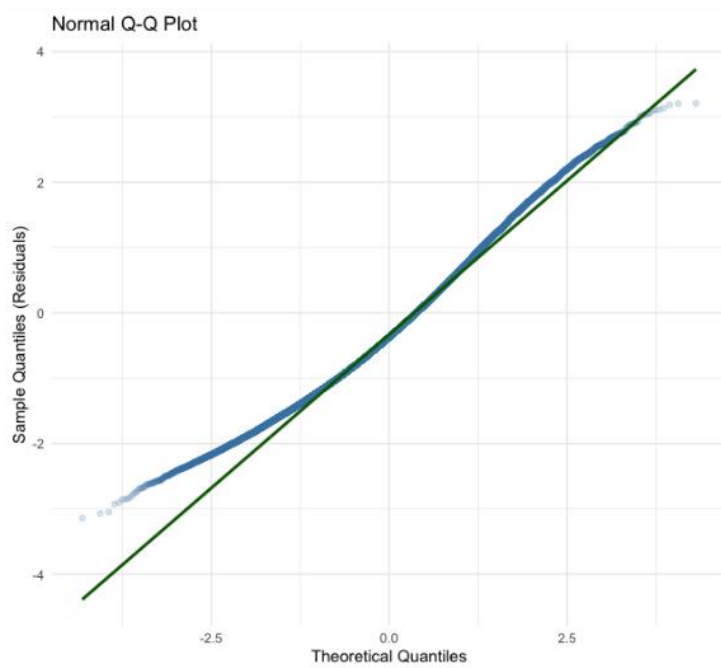
Homoscedasticity and linearity of relationship between predictors and outcome variable



Note. This figure shows residual diagnostics for the Gamma generalized linear mixed model with a log-link function. The deviance residuals are plotted against the linear predictor values. The horizontal alignment of the red line demonstrates consistent linearity across the range of fitted values with no evidence of systematic misspecification.

Figure B2

Normality of residuals



Note. The sample quantiles of the model's deviance residuals are plotted against theoretical normal quantiles. The alignment of the empirical data along the reference line (green) indicates an acceptable distribution of errors.